

Project Workflow: Week 6

Phase: Items, Inventory & Torch Mechanic

1. Objective:

The goal of Week 6 was to introduce core gameplay mechanics and survival dynamics to the island exploration. This included:

- Implementing coins, clue scrolls, and treasure chest logic (Items.java).
- Adding an inventory toggle system using the 'T' key.
- Implementing a torch fuel countdown system with darkness masking and a game-over condition when the torch expires.

2. Detailed Task Breakdown

Implementing Coins, Clue Scrolls, and Treasure Chest Logic (Items.java) Created the static outer Items class along with an abstract inner Item class defining shared properties such as grid position, item type, collection state, and rendering colors. Built specific subclasses (Coin, ClueScroll, TorchFuelPickup, and TreasureChest). Updated GameMap to randomly distribute these items across walkable island tiles and implemented collision handling in GamePanel to allow the player to pick up items upon stepping on them.

Implementing Inventory Toggle System Integrated state-based UI rendering for an interactive inventory system. Added key handling for the 'T' key inside GamePanel to switch the game state between PLAYING and INVENTORY. When in the inventory state, the rendering engine displays a centered overlay listing all collected clue scrolls and their associated riddles without resetting game progress.

Implementing Torch Fuel Countdown & Darkness/Game-Over Condition Added a torchFuel variable to the Player model and programmed a continuous fuel depletion rate linked to the 60 FPS game loop. Created a Graphics2D darkness mask using java.awt.geom.Area and Ellipse2D to restrict visual visibility around the player based on the remaining fuel level. Implemented game-over handling so that when fuel completely depletes, the game transitions to a GAME_OVER overlay screen and prompts the user to restart.

3. Deliverables

- **Items Logic (Items.java):** Working implementation of coins, clue scrolls, and treasure chest logic.
- **Inventory Toggle System:** Functioning inventory overlay toggled via the 'T' key.
- **Torch Fuel Countdown System:** Continuous fuel drain mechanics and visual fuel tracking.

- **Darkness & Game-Over Condition:** Ambient darkness masking and state transition triggered when torch fuel expires.

4. Tools & Resources

- **Development Environment:** Java JDK 8+, IntelliJ IDEA / VS Code / Eclipse.
- **Graphics & Geometry APIs:** `java.awt.Graphics2D`, `java.awt.geom.Area`, `java.awt.geom.Ellipse2D` for ambient lighting and darkness blending.
- **Reference Materials:** Java Swing Timer documentation for frame-based resource decay and Object-Oriented Programming (OOP) inheritance guidelines for item structures.